Table 16.4: Comparison between field-effect transistor and bipolar junction transistor

S.No	*Field-Effect Transistor (FET)*	*Bipolar junction transistor (BJT)*
1.	It is a unipolar device. *i.e.*, current in the device is carried either by electrons or holes.	It is a bipolar device. *i.e.* current in the device is carried by both electrons and holes.
2.	It is a voltage-controlled device *i.e.*, voltage at the gate (or drain) terminal controls the amount of current flowing through the device.	It is a current-controlled device. *i.e.* the base current controls the amount of collector current.
3.	Its input resistance is very high and is of the order of several megaohms.	Its input resistance is very low as compared to FET and is of the order of few kilohms.

4.	It has a negative temperature coefficient at high current levels. It means that current decreases as the temperature increases. This characteristic prevents the FET from thermal breakdown.	It has a positive temperature coefficient at high current level. It means that collector current increases with the increase in temperature. This characteristic leads the BJT to thermal breakdown.
5.	It does not suffer from minority-carrier storage effects and therefore has higher switching speeds and cut-off frequences.	It suffers from minority carrier storage effects and therefore has lower switching speed and cut-off frequencies than that of FET's.
6.	It is less noisy than a BJT or vacuum tube and is thus more suitable as an input amplifier for low-level signals. It is used extensively in high fidelity frequency modulated receivers.	It is comparatively more noisy than a field-effect transistor.
7.	It is much simpler to fabricate as a integrated circuit (IC) and occupies a less space on IC chip than that of BJT.	It is comparatively difficult to fabricate as an integrated circuit and occupies more space on IC chip than that of FET.

16.14. MOSFETs

The MOSFET is an abbreviation for metal-oxide semiconductor field-effect transistor. Like JFET, it has a source, gate and drain. However, unlike JFET, the gate of a MOSFET is insulated from the channel. Because of this, the MOSFET is sometimes known as an IGFET which stands for insulated-gate field effect transistor.

16.15. Types of MOSFET

There are two basic types of MOSFETs; depletion-type MOSFETs and enhancement-type MOSFETs. The depletion type MOSFETs are also called D-type MOSFETs and enhancement-type MOSFETs. as E-type MOSFETs. The primary difference between the two types of MOSFETs is the difference between the constructions. Both the D-type and E-type MOSFETs are disscussed one by one in the following pages.

16.16. Depletion-Type MOSFET

Fig. 16.15 shows the basic structure of an N-channel depletion type MOSFET. It consists of a conducting bar of N-type material with an insulated gate on the left and P-region on the right. Free electrons can flow from source to drain through the N-type material. The P-region is called substrate (or body). It physically reduces the conducting path to a narrow channel. A thin layer of silicon dioxide is deposited on the left side of the channel. This layer insulates the gate from the channel. Because of this, a negligible gate current flows even when the gate voltage is positive. It will be interesting to know that a PN junction, which exists in a JFET, has been eliminated in the MOSFET.

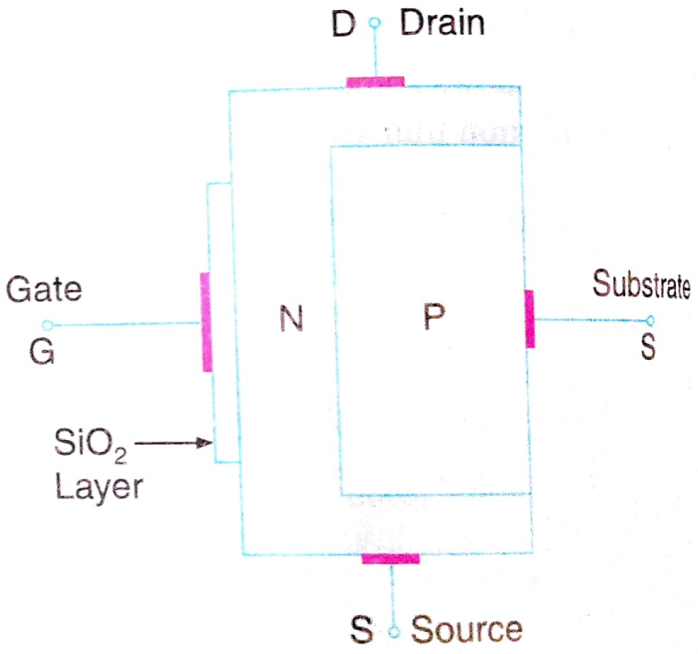


Fig. 16.15. Basic structure of an N-channel depletion-type MOSFET.

16.17. Working of a Depletion-Type MOSFET

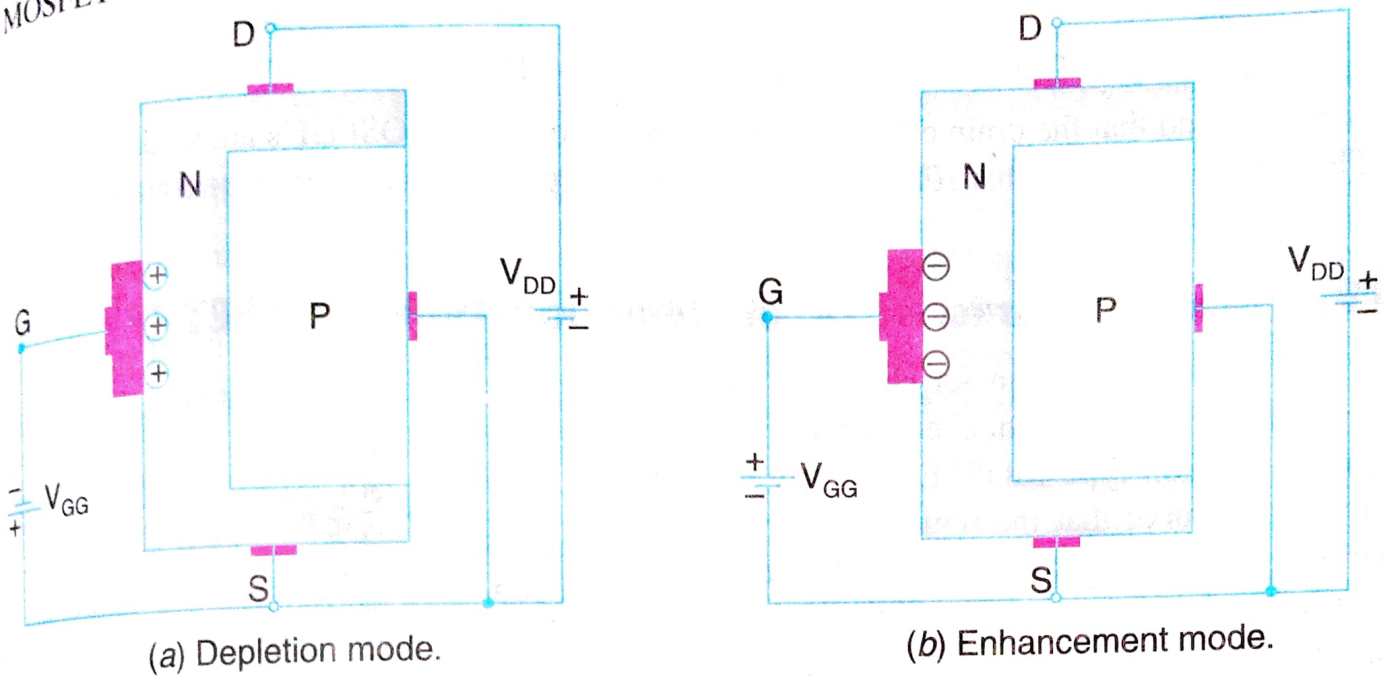
The depletion-type MOSFET can be operated in two different modes as given below:

1. *Depletion mode.* The device operates in this mode, when the gate voltage in negative.
2. *Enhancement mode.* The device operates in this mode, when the gate voltage is positive.

Since the depletion-type MOSFET can be operated in either depletion or enhancement mode, therefore this device is commonly known as depletion-enhancement (DE) type MOSFET. The working of a MOSFET may be explained easily, if we visualize the entire structure of the device as a parallel plate capacitor. One of the plates is formed by the gate and the other by the semiconductor channel. The plates are separated by a dielectric (SiO_2 layer). We know that if one plate of a capacitor is made negative, it induces a positive charge on the opposite plate and vice versa. This principle is used below in explaining the working of MOSFET's in the depletion and enhancement modes.

1. *Depletion mode.* Fig. 16.16 (*a*) shows a MOSFET with a negative gate-to-source voltage. The negative voltage, on the gate, induces a positive charge in the channel. Because of this, free electrons in the vicinity of positive charge are repelled away in the channel. As a result of this, the channel is depleted of free electrons. This reduces the number of free electrons (which constitute the drain current) passing through the channel. Thus as the value of negative gate-to-source voltage is increased, the value of drain voltage, called $V_{GS(off)}$, the channel is totally depleted of free electrons and therefore the drain current reduces to zero. Thus with the negative gate voltage, the operation of MOSFET is similar to that of a JFET

MOSFET is similar to that of a JFET



(*a*) Depletion mode. (*b*) Enhancement mode.

Fig. 16.16. Working of a depletion-type MOSFET.

It is evident from the above discussion that negative gate voltage depletes the channel of free electrons. It is due to this fact that the working of a MOSFET, with a negative gate voltage, is called depletion mode.

2. *Enhancement mode.* Figure 16.16 (*b*) shows a MOSFET with a positive gate-to-source voltage. The positive gate voltage increases the number of free electrons passing through the channel. The greater the gate voltage, greater is the number of free electrons passing through the channel. This increases *i.e.*, enhances the conducting of the channel. Because of this fact, positive gate operation is called enhancement mode.

16.18. Drain Characteristic of Depletion-Type MOSFET

Fig. 16.17 shows the drain characteristic for the N-channel depletion-type MOSFET in the common source configuration. These curves are plotted for both negative and positive values of gate-to-source voltage (V_{GS}). The curves shown above the curve for $V_{GS} = 0$ have a positive zero whereas those below it have a negative value of V_{GS}. When V_{GS} is zero and negative, the MOSFET operates in the depletion-mode. On the other hand, if VGS is zero and positive, the MOSFET operates in the enhancement-mode.

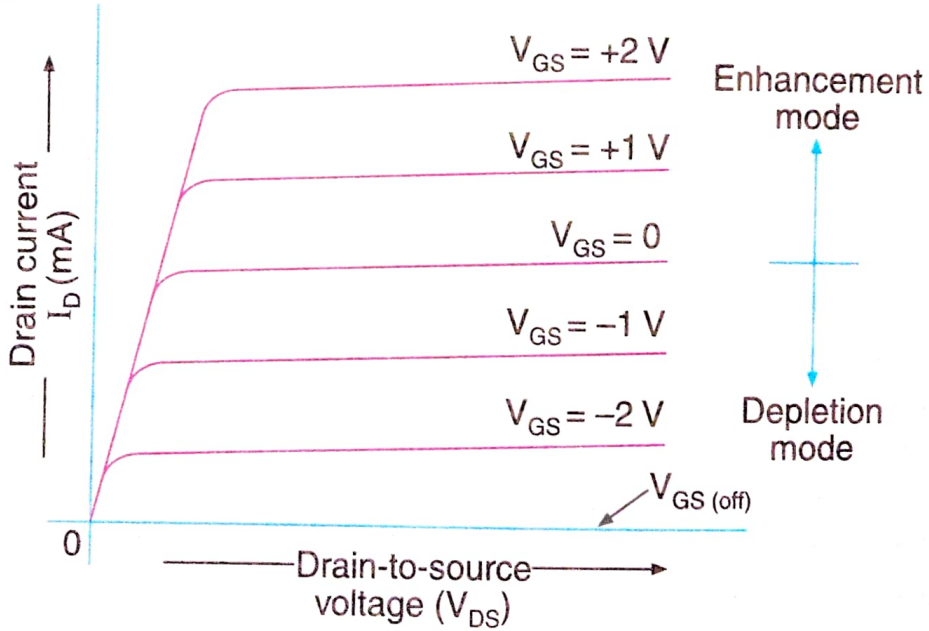


Fig. 16.17. Drain characteristics of N-channel depletion-type MOSFET.

It may be noted that the drain characteristics of depletion-type MOSFET's are similar to that of JFET. The only difference is that JFET does not operate for positive values of gate-to-source voltage (V_{GS}).

16.19. Transfer Characteristic of Depletion-Type MOSFET

Fig. 16.18 shows the transfer characteristic (also called transconductance curve) for an N-channel depletion-type MOSFET. It may be noted from this curve that the region *AB* of the characteristic is similar to that of JFET. But here, this curve extends for the positive values of gate-to-source voltage (V_{GS}) also. The value I_{DSS} represents the current from drain-to-source with $V_{GS} = 0$. The drain current at any point along the transfer characteristic (*i.e.*, the curve *ABC*) is given by the relation,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

It may be noted that even if $V_{GS} = 0$, the device has a drain current equal to I_{DSS}. Due to this fact, it is called normally-ON MOSFET.

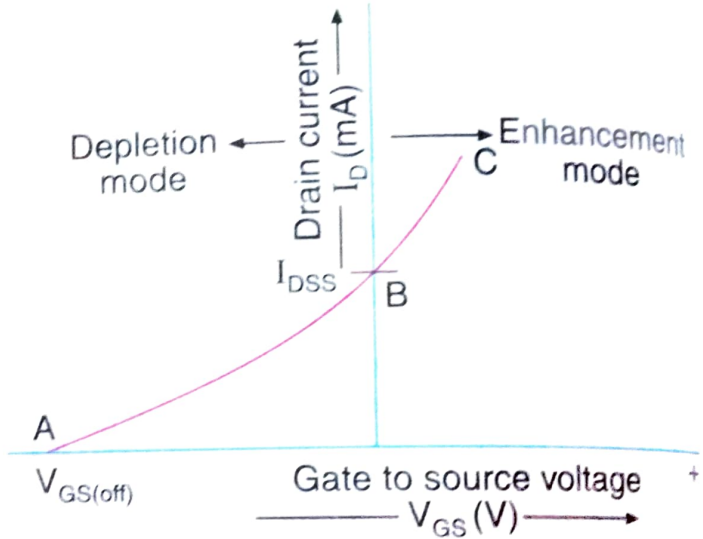


Fig. 16.18. Transfer characteristics of a N- channel depletion-type MOSFET.